🍕 Complete Data Summary & Strategic Recommendations

(Based on Dashboard + Raw Data Files)

1️⃣ Data Model Understanding (Very Important for Client)

Your dataset is **well-structured and clean**:

File	Purpose
orders.csv	Order date & time (customer demand behavior)
order_details.csv	Line-level order items & quantity
pizzas.csv	Price & size of each pizza
pizza_types.csv	Category, name & ingredients

👉 This is a **perfect star-schema style structure**, ideal for analytics & BI tools like Power BI / Tableau.

2️⃣ Business KPI Summary (Verified from Raw Data)

🔑 Core Performance Metrics

- **Total Orders:** 21,350
- **Total Pizzas Sold:** 49,574
- **Total Revenue:** ₹8,17,860.05
- **Avg Pizzas per Order: 2.32**

🧠 What This Means

- Customers usually order **more than 1 pizza**, confirming **strong bundling behavior**.
- Revenue is **volume-driven**, not dependent on a few high-value orders.
- Business shows **stable, repeatable demand** → very healthy sign.

3️⃣ Category-Level Performance (Revenue Based)

Category	Revenue (₹)	Insight
Classic	2,20,053	Highest performing
Supreme	2,08,197	Strong premium appeal
Chicken	1,95,919	Consistent demand
Veggie	1,93,690	Almost equal to Chicken

🧠 **Key Insight**

- Revenue distribution is **balanced**.
- No category is dependent on niche demand.
- **Classic pizzas are the backbone** of the business.

🎯 **Recommendation**

👉 Use **Classic pizzas as anchor products**
👉 Cross-sell Supreme / Chicken as upgrades

4️⃣ **Size-Level Performance (Critical Business Insight)**

Size	Revenue (₹)	Contribution
Large (L)	3,75,318	🔥 Dominant
Medium (M)	2,49,382	Strong
Small (S)	1,78,076	Moderate
XL	14,076	Very Low
XXL	1,006	Negligible

🧠 **Key Insight (Very Strong)**

- **Large + Medium = ~76% of total revenue**
- XL & XXL together contribute **< 2%**

🚨 **Strong Recommendation**

✅ Optimize inventory for **L & M only**
❌ Reduce XL & XXL production (high waste risk)
🎯 Run "Upgrade to Large at ₹X" offers → instant revenue lift

5️⃣ Top Performing Pizzas (Revenue Heroes)

Pizza Name	Revenue (₹)
Thai Chicken Pizza	43,434
Barbecue Chicken Pizza	42,768
California Chicken Pizza	41,409
Classic Deluxe Pizza	38,180
Spicy Italian Pizza	34,831

🧠 **Insight**

- **Chicken-based pizzas dominate top ranks**
- Customers prefer **bold flavors + protein-rich options**

🎯 **Recommendation**

👉 Highlight these pizzas:

- Homepage
- Combo deals
- Featured banners
- Recommendation engine

6️⃣ Bottom Performing Pizzas (Confirmed by Data)

Pizza Name	Revenue (₹)
Brie Carre Pizza	11,588
Green Garden Pizza	13,955
Mediterranean Pizza	15,360
Spinach Pesto Pizza	15,596
Spinach Supreme Pizza	15,277

🧠 **Insight (Critical)**

- Mostly **Veggie + premium ingredient pizzas**
- Indicates:
  - Taste mismatch OR
  - Pricing resistance OR
  - Poor visibility

🚨 **Strong Business Decision**

Choose ONE:

1. Reprice (budget-friendly)
2. Reposition (health / diet category)
3. Remove from menu (menu optimization)

Clients LOVE this kind of clarity.

7️⃣ **Time-Based & Day-Based Confirmation**

Your **dashboard peak hours & days are 100% consistent with raw data**:

- 🔥 Peak Hours: **12–3 PM & 6–9 PM**
- 🔥 Peak Days: **Friday & Saturday**
- ⬇️ Low Demand: Morning hours & Sundays

🎯 **Operational Recommendation**

- Increase kitchen & delivery staffing during peak slots
- Introduce **Sunday-only family combos**
- Launch **morning discount experiments**

8️⃣ **Executive-Level Strong Recommendations (Client-Ready)**

📌 **Revenue Growth**

- Push **Large size upgrades**
- Bundle **Classic + Chicken combos**
- Promote top 5 pizzas aggressively

📌 **Cost Optimization**

- Reduce XL / XXL inventory
- Rationalize low-selling veggie pizzas

📌 **Marketing Strategy**

- Time-based offers (morning & Sunday)
- Highlight “Most Loved Pizzas” using real data
- Loyalty offers for repeat customers

📌 **Menu Strategy**

- Keep menu focused, not crowded
- Data-driven removal improves margins

9️⃣ **Final Client Conclusion (Use This Line)**

“The data clearly shows a healthy, volume-driven pizza business with strong customer preferences toward Large-sized, Classic and Chicken pizzas. By optimizing inventory, focusing promotions on top-performing products, and rationalizing underperforming items, the business can significantly improve profitability without increasing operational complexity.”